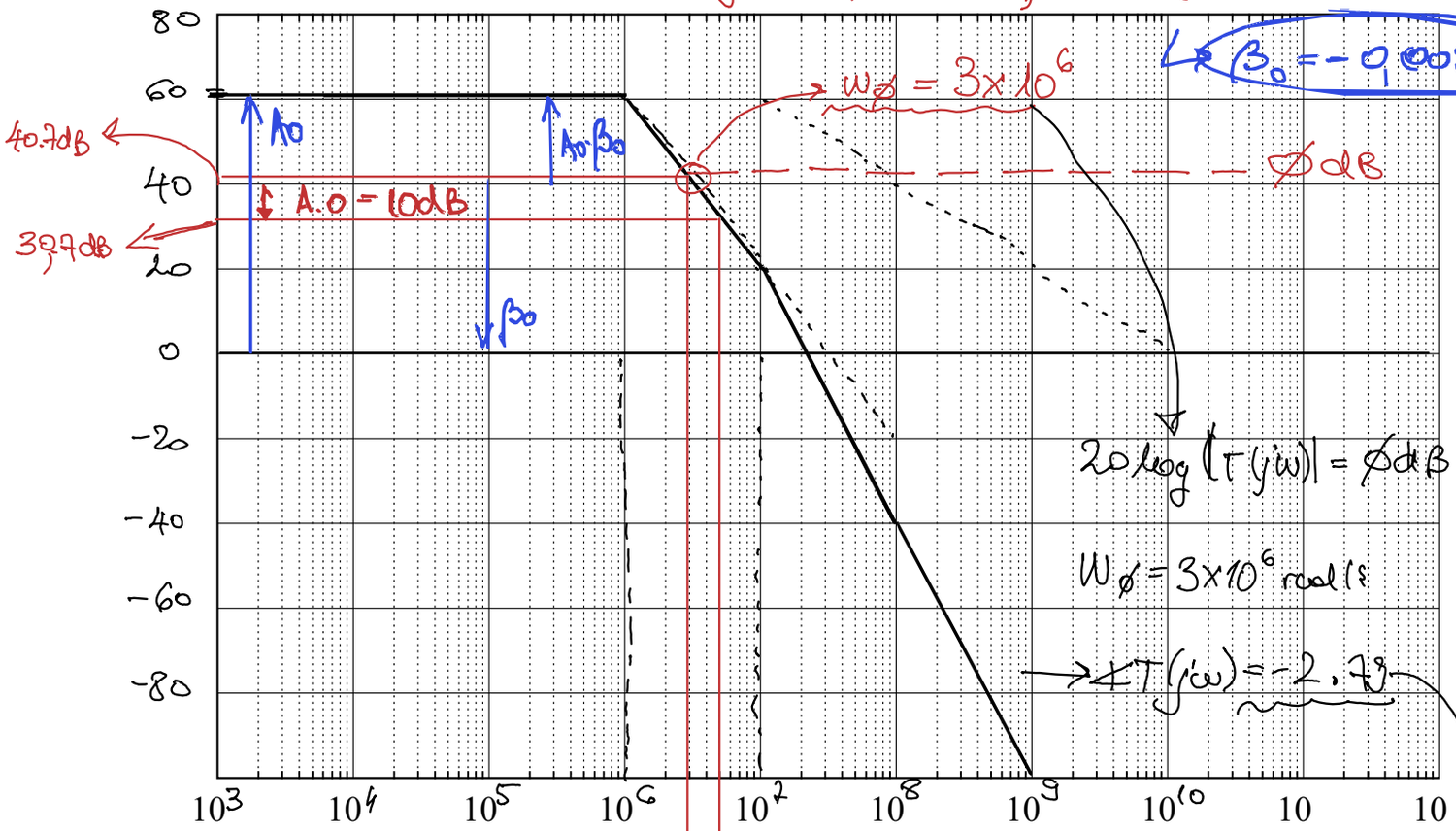
$20\log(A_0\beta_0) = 19.3\,dB$ $\quad 20\log(\beta_0) = 40.7\,dB$

$\beta_0 = -0.0091$



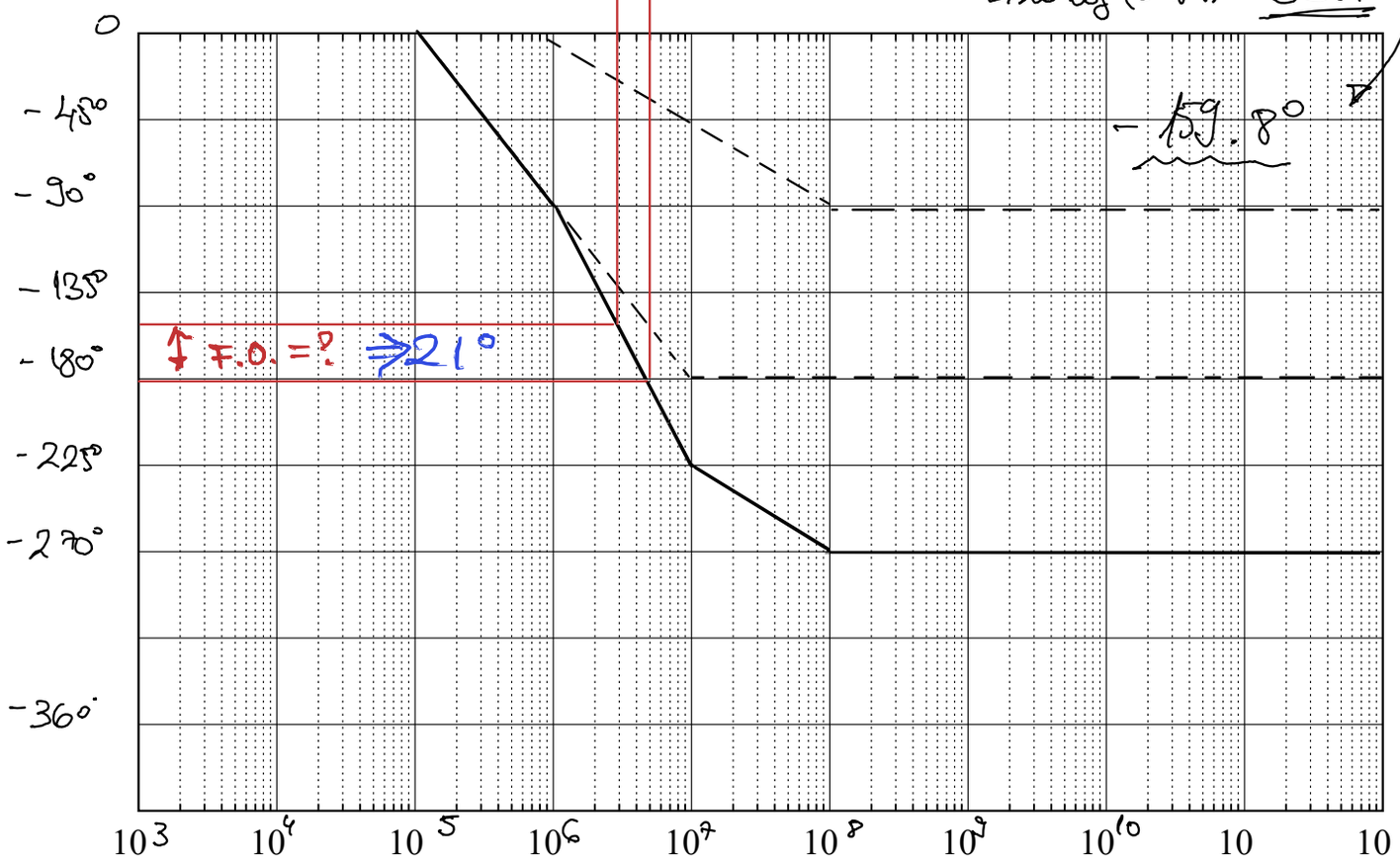
$20\log|T(j\omega)| = \phi\,dB$

$\omega_\phi = 3\times10^6$ rad/s

$\rightarrow \angle T(j\omega) = -2.7°$

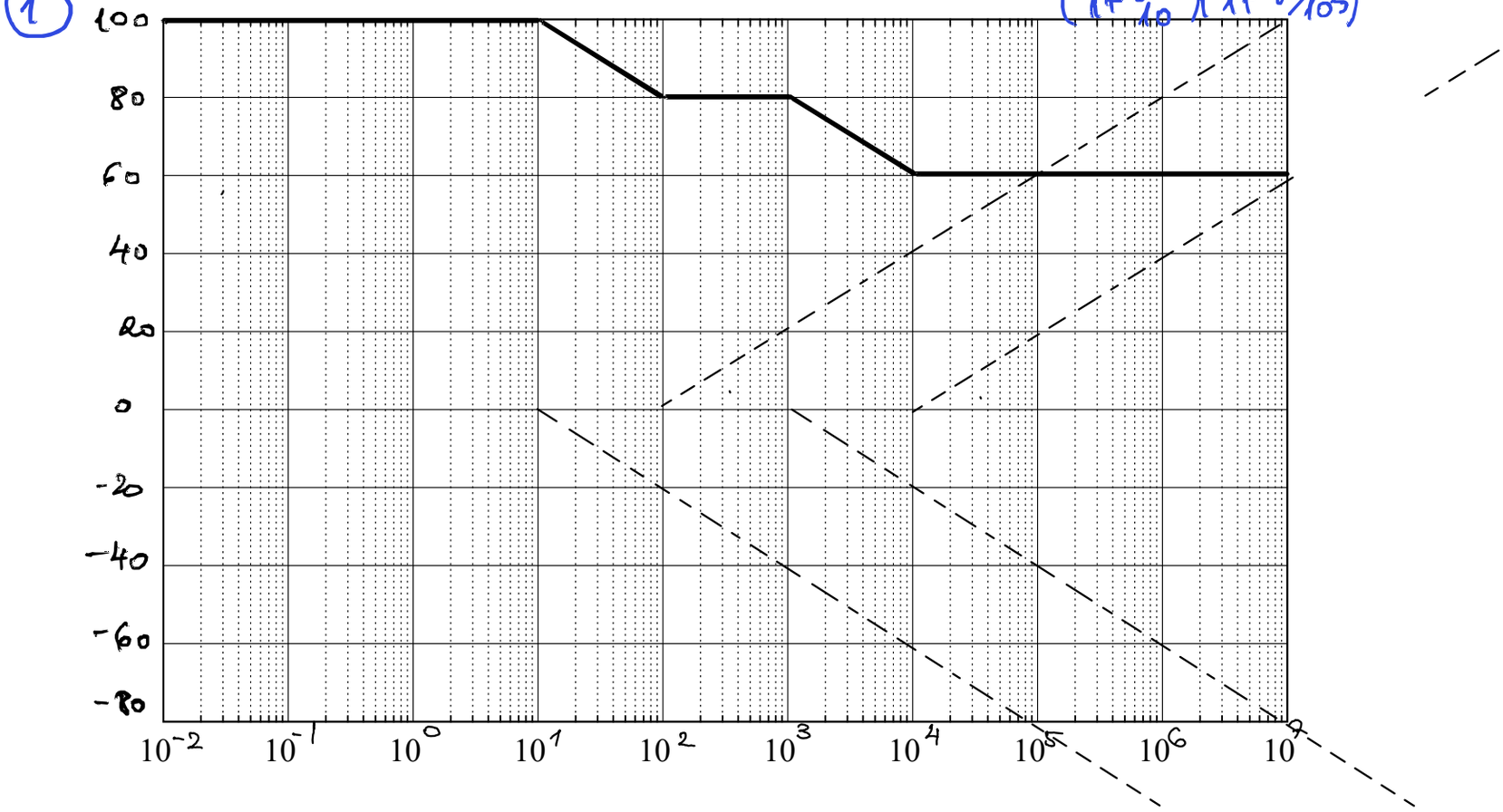
$\omega_{180} = 5\times10^6 \rightarrow |T(j\omega)| = 34.4$

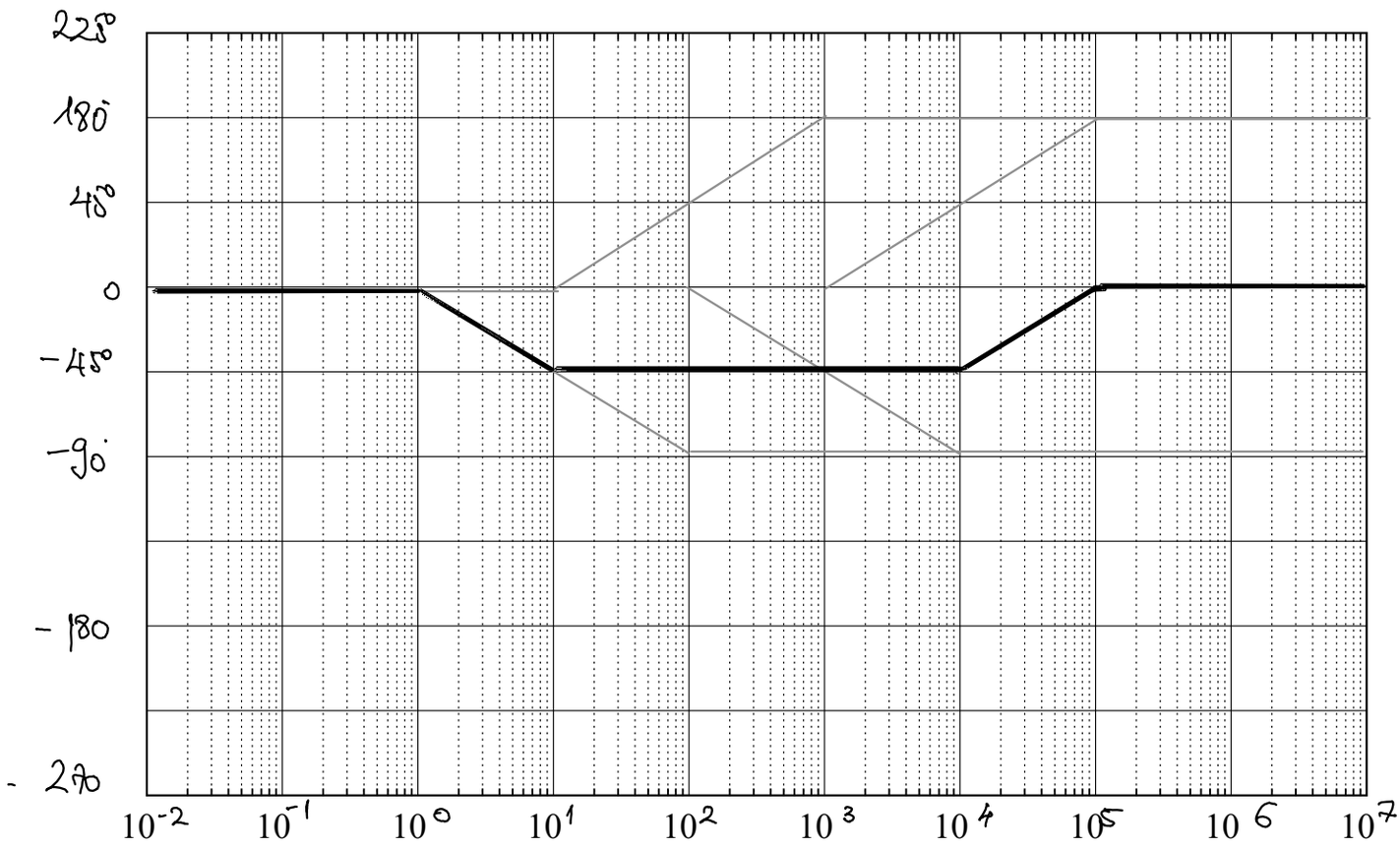
$\hookrightarrow 20\log(34.4) = 30.7$



Zadaci 7.2. $A(j\omega) = 10^5 \dfrac{\left(1+\frac{j\omega}{10^2}\right)\left(1+\frac{j\omega}{10^4}\right)}{\left(1+\frac{j\omega}{10}\right)\left(1+\frac{j\omega}{10^3}\right)}$

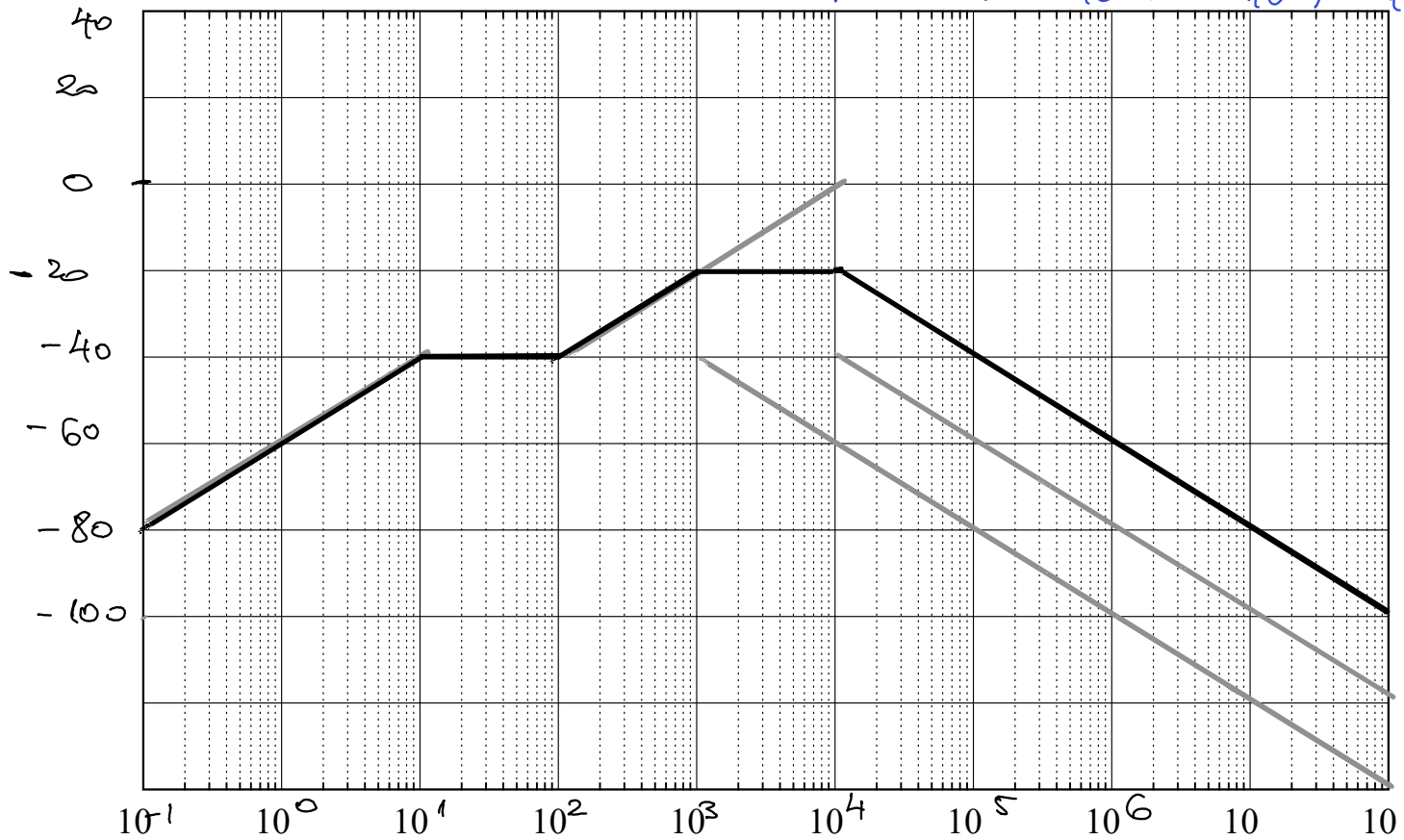
1

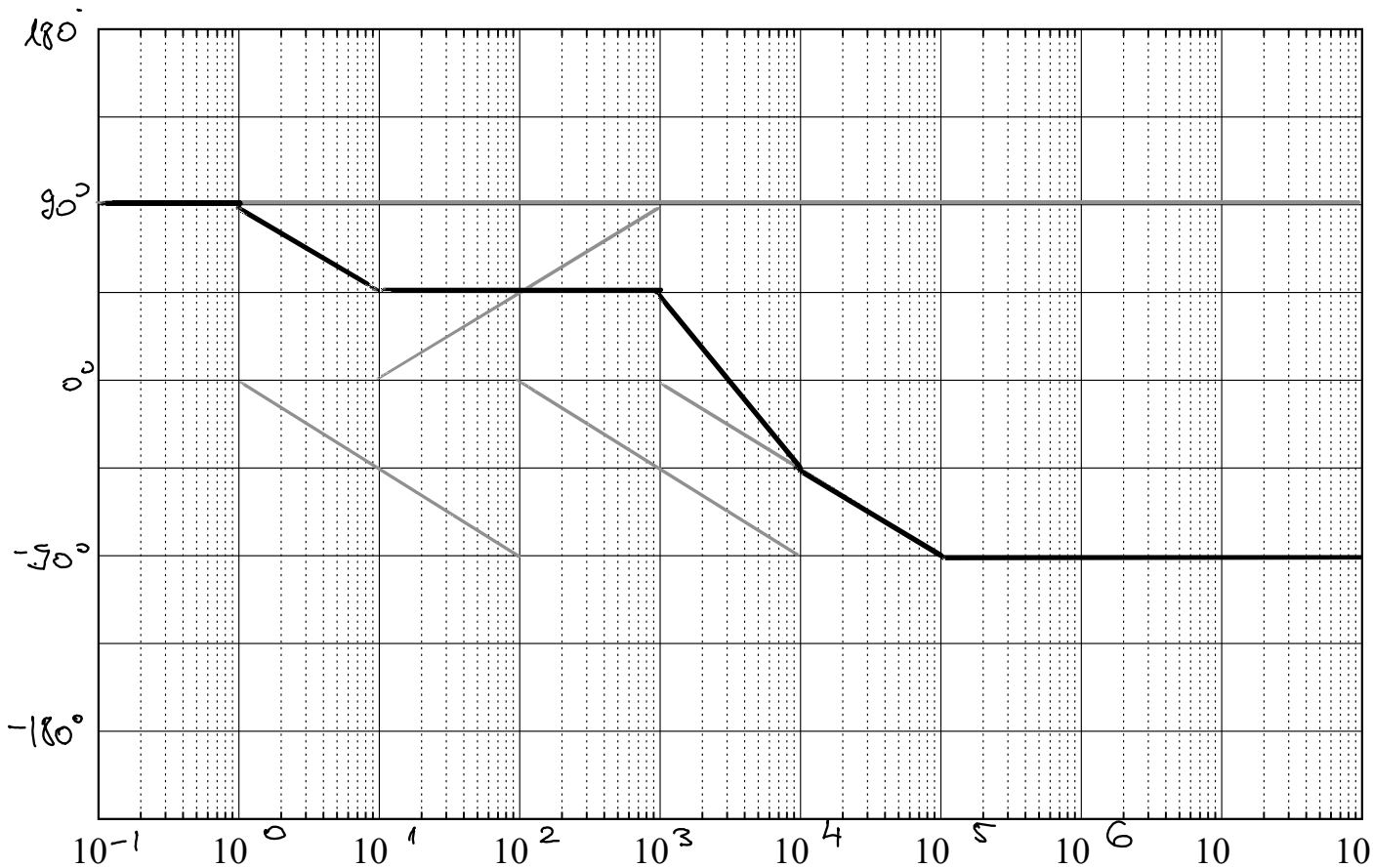




2

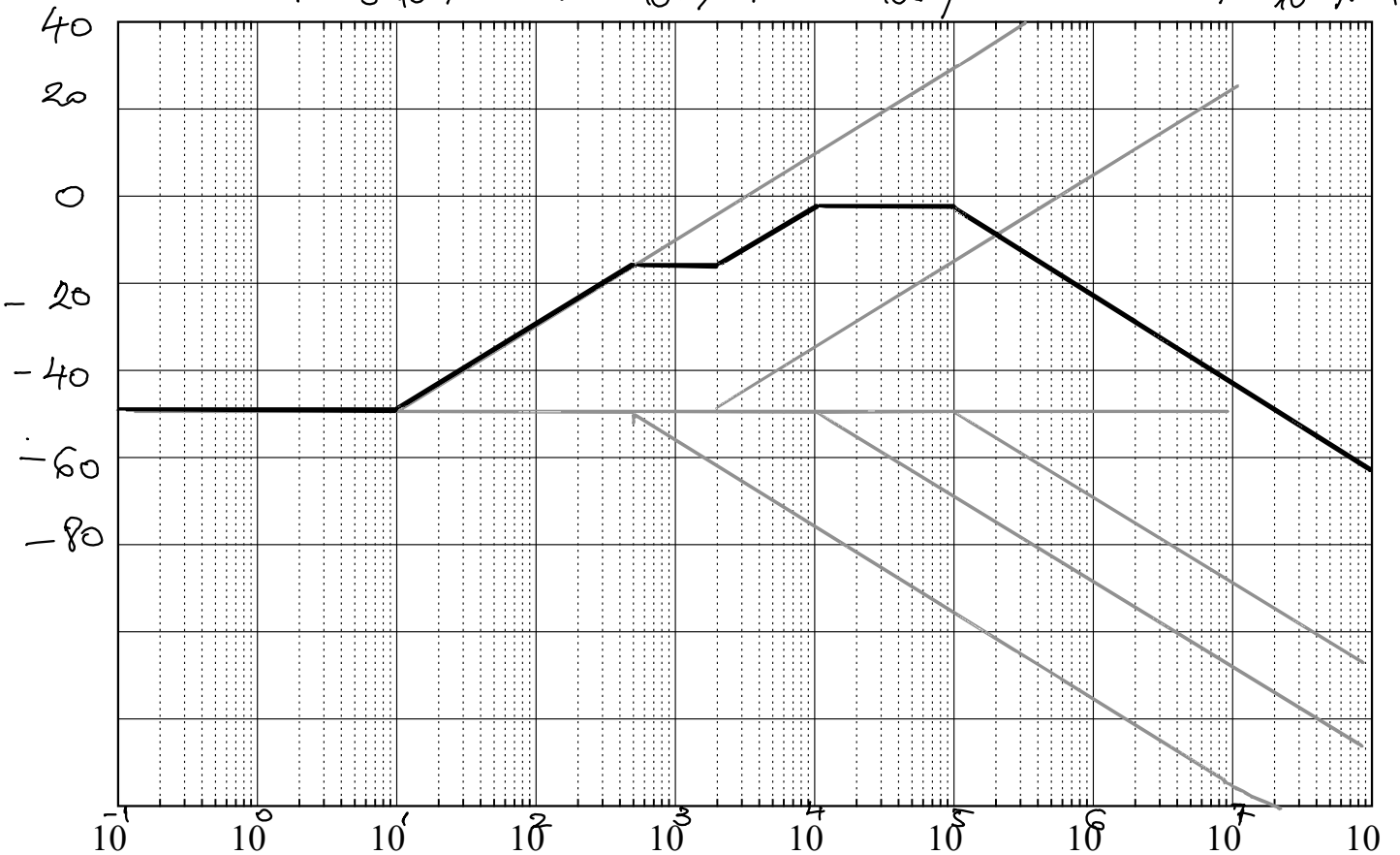
$$\frac{10^{3}\,10^{2}\cdot j\omega\left(1+\frac{j\omega}{10^{2}}\right)}{10^{1}\left(1+\frac{j\omega}{10}\right)\cdot 10^{3}\left(1+\frac{j\omega}{10^{3}}\right)\cdot 10^{4}\left(1+\frac{j\omega}{10^{4}}\right)} = 10^{-2}\,\frac{\frac{j\omega}{10^{1}}\left(1+\frac{j\omega}{10^{2}}\right)}{\left(1+\frac{j\omega}{10}\right)\left(1+\frac{j\omega}{10^{3}}\right)\left(1+\frac{j\omega}{10^{4}}\right)}$$

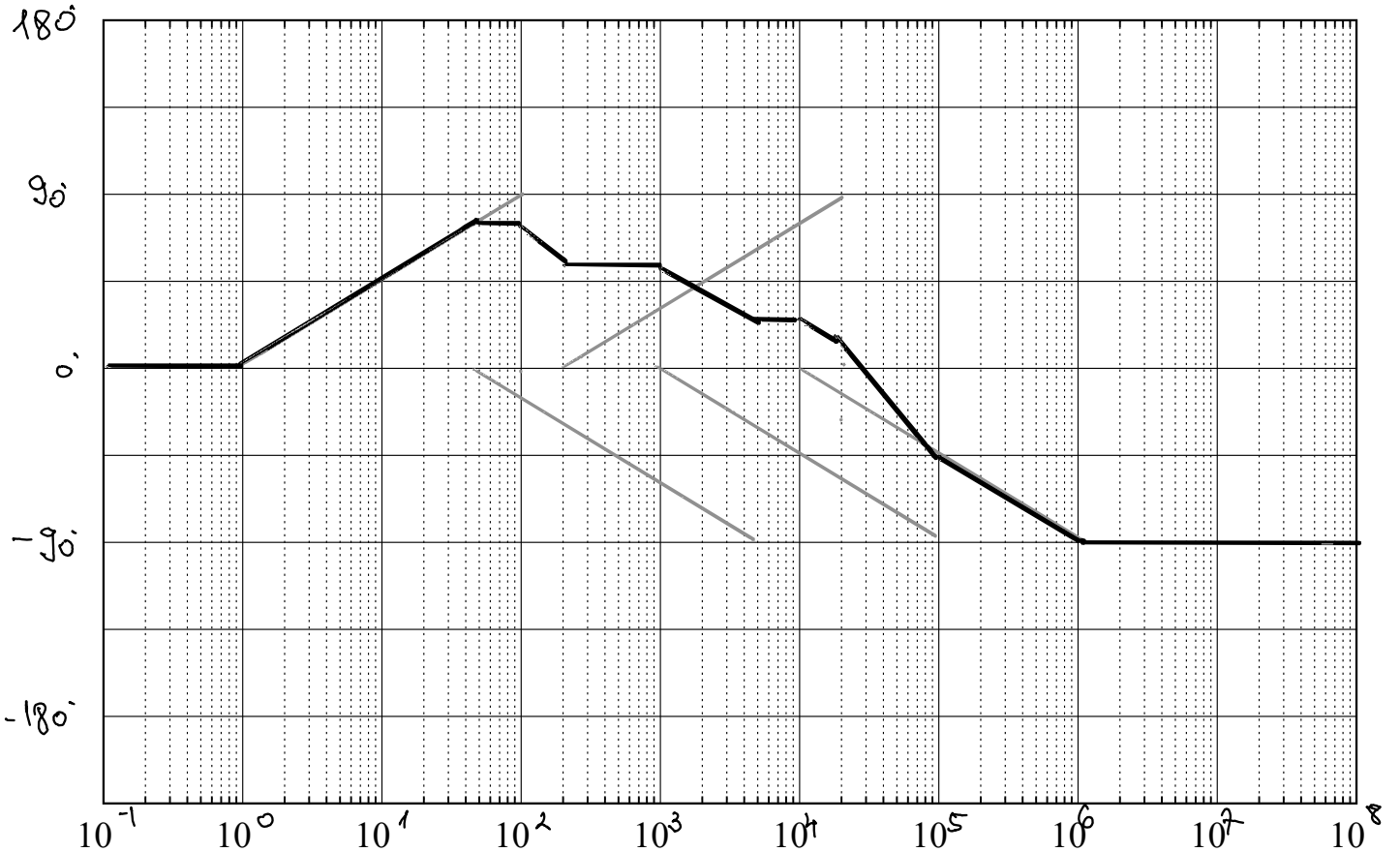


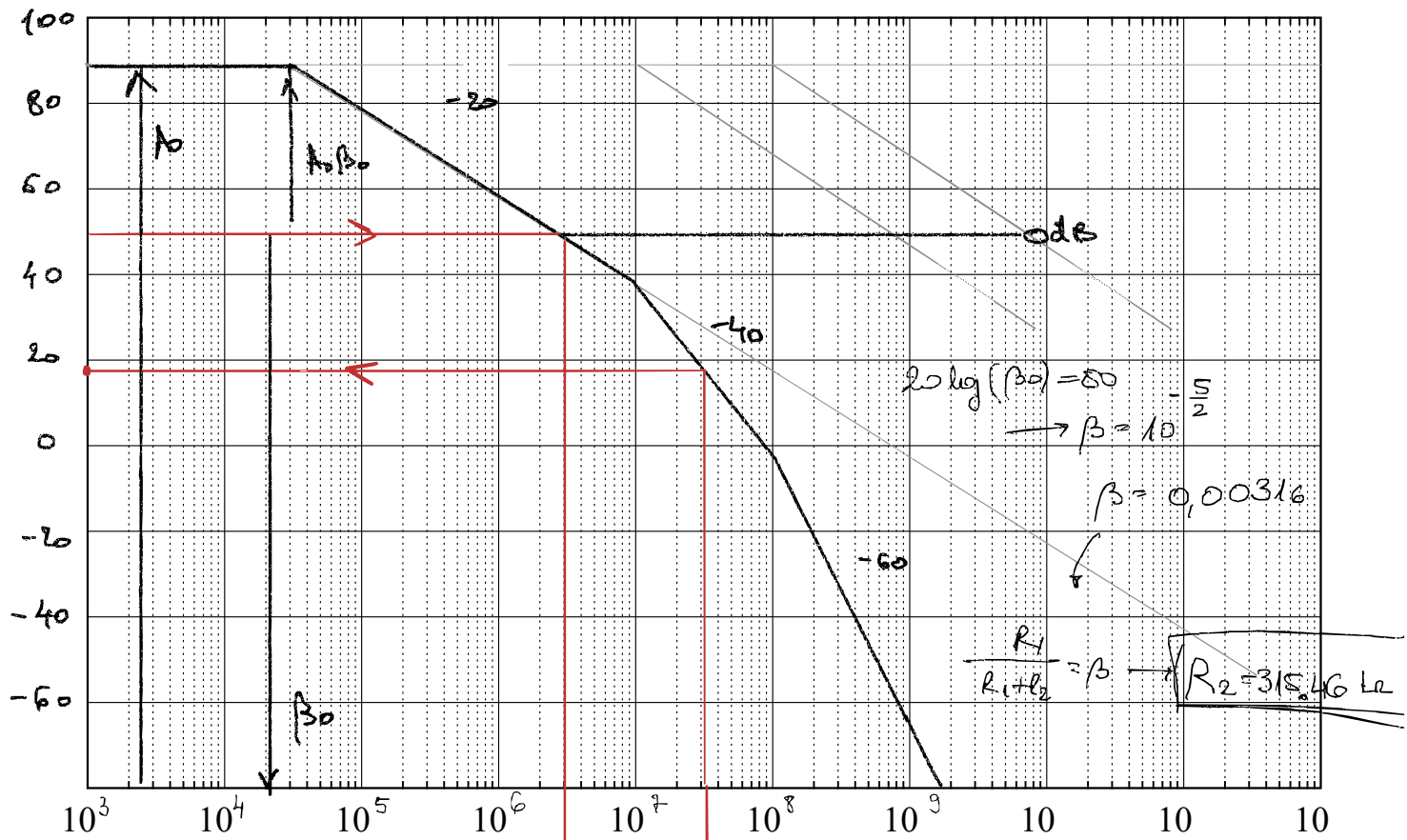


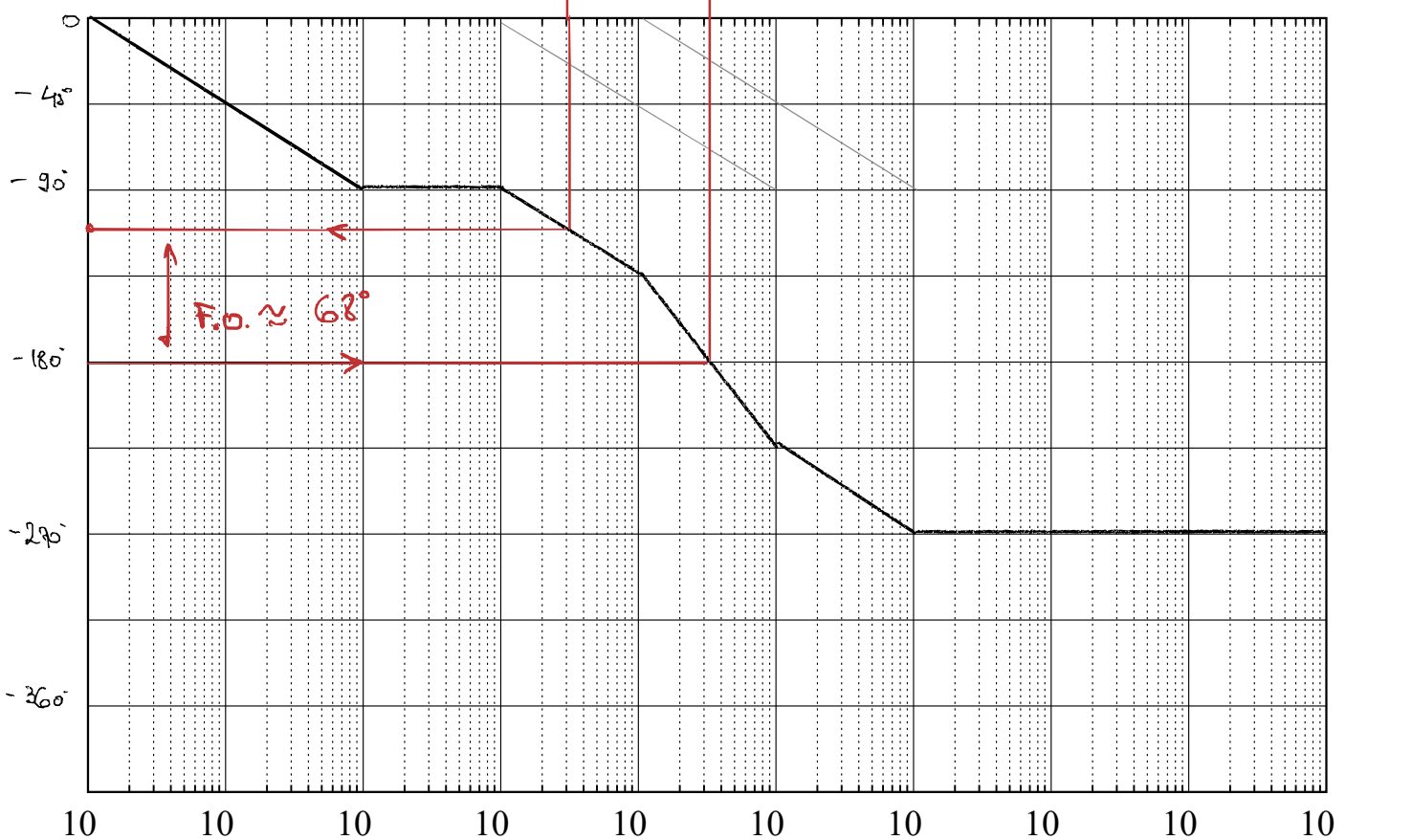
3.

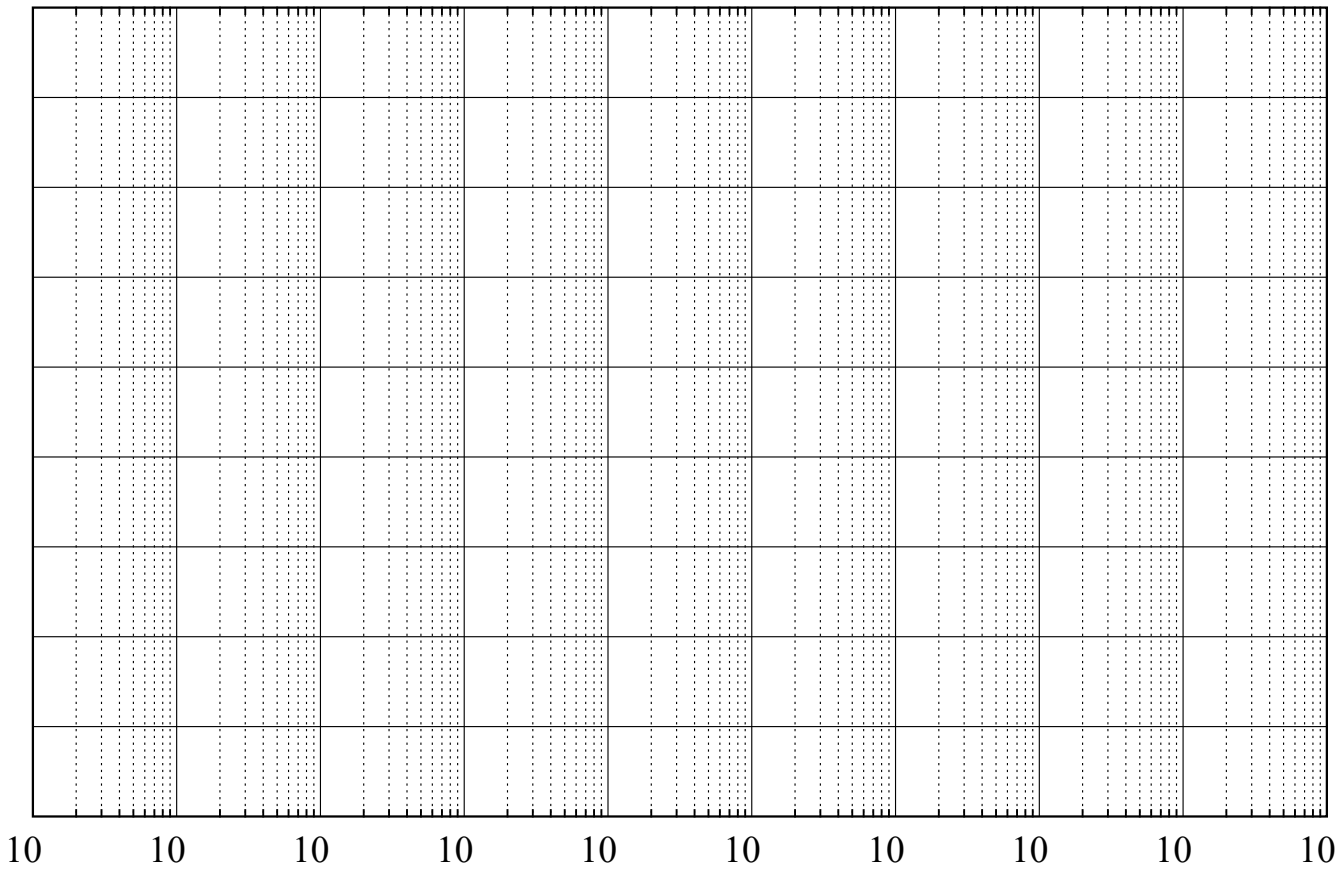
$$\cancel{10^5}\cdot\frac{\cancel{10}\left(1+\frac{j\omega}{10}\right)\cdot 2\cdot\cancel{10^3}\left(1+\frac{j\omega}{2\cdot 10^3}\right)}{5\cdot 10^2\left(1+\frac{j\omega}{5\cdot 10^2}\right)\cdot\cancel{10^4}\left(1+\frac{j\omega}{10^4}\right)\cdot\cancel{10^5}\left(1+\frac{j\omega}{10^5}\right)}=\frac{2}{5}\cdot 10^{-2}\cdot\frac{\left(1+\frac{j\omega}{10}\right)\left(1+\frac{j\omega}{2\cdot 10^3}\right)}{\left(1+\frac{j\omega}{5\cdot 10^2}\right)\left(1+\frac{j\omega}{10^4}\right)\left(1+\frac{j\omega}{10^5}\right)}$$

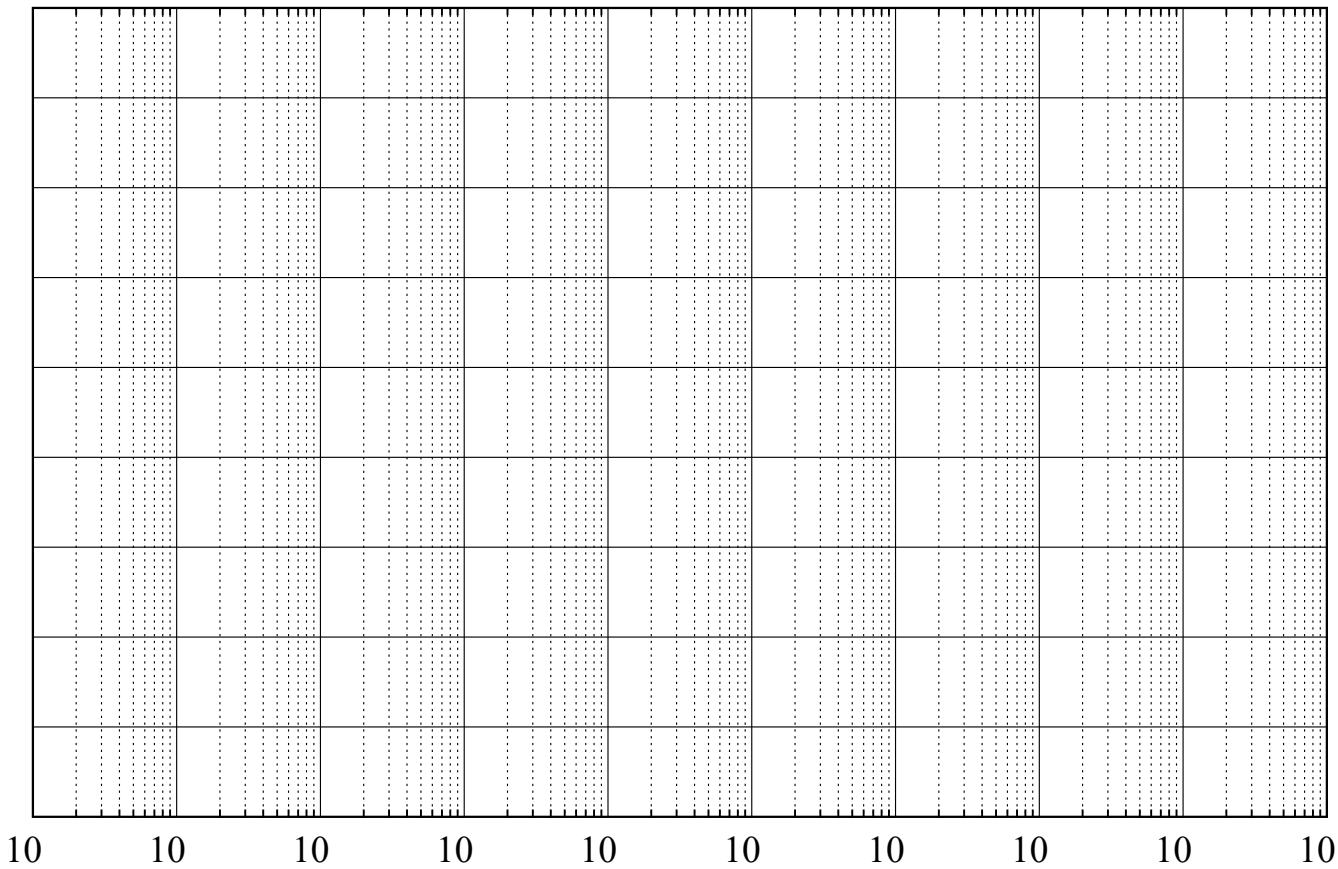


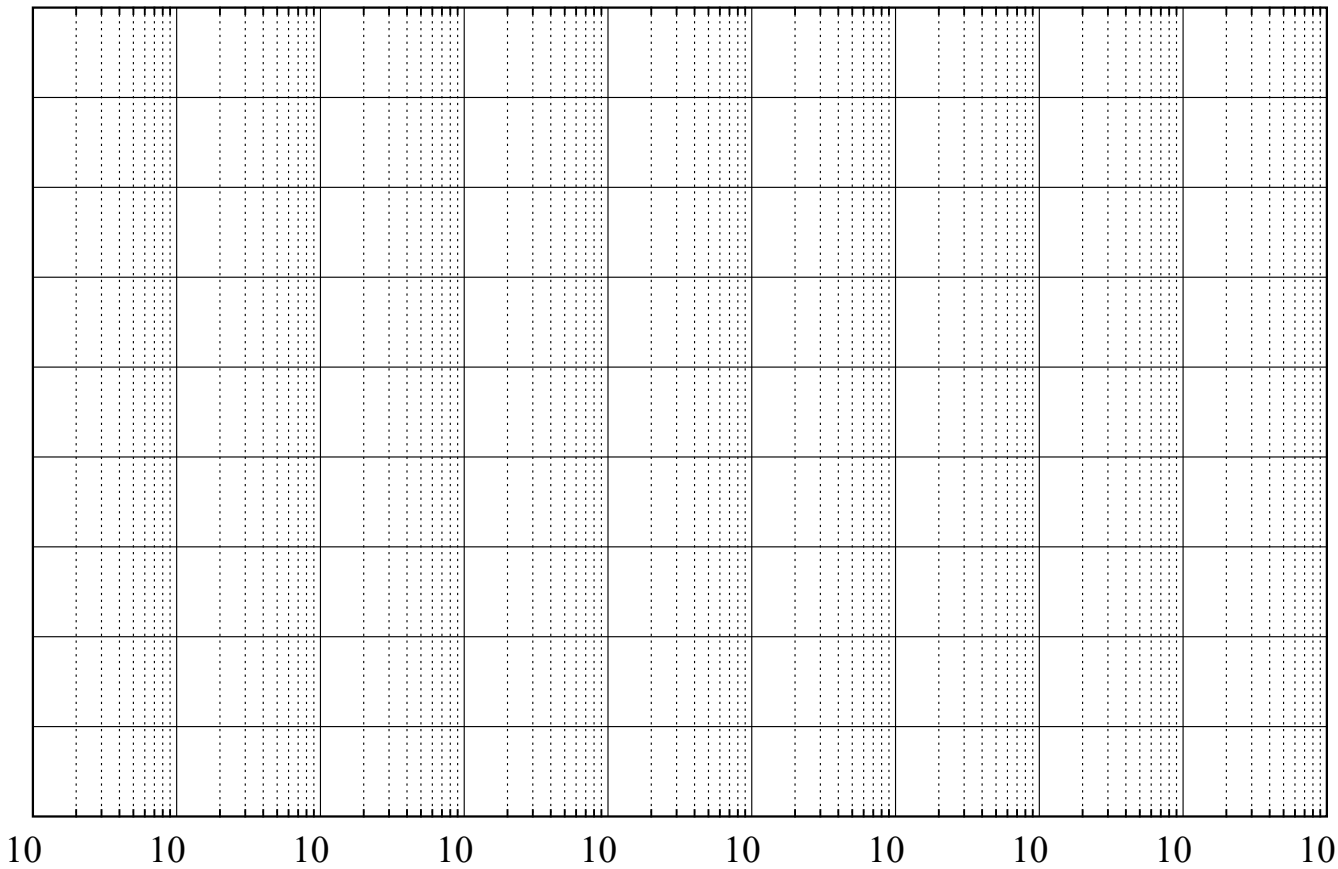


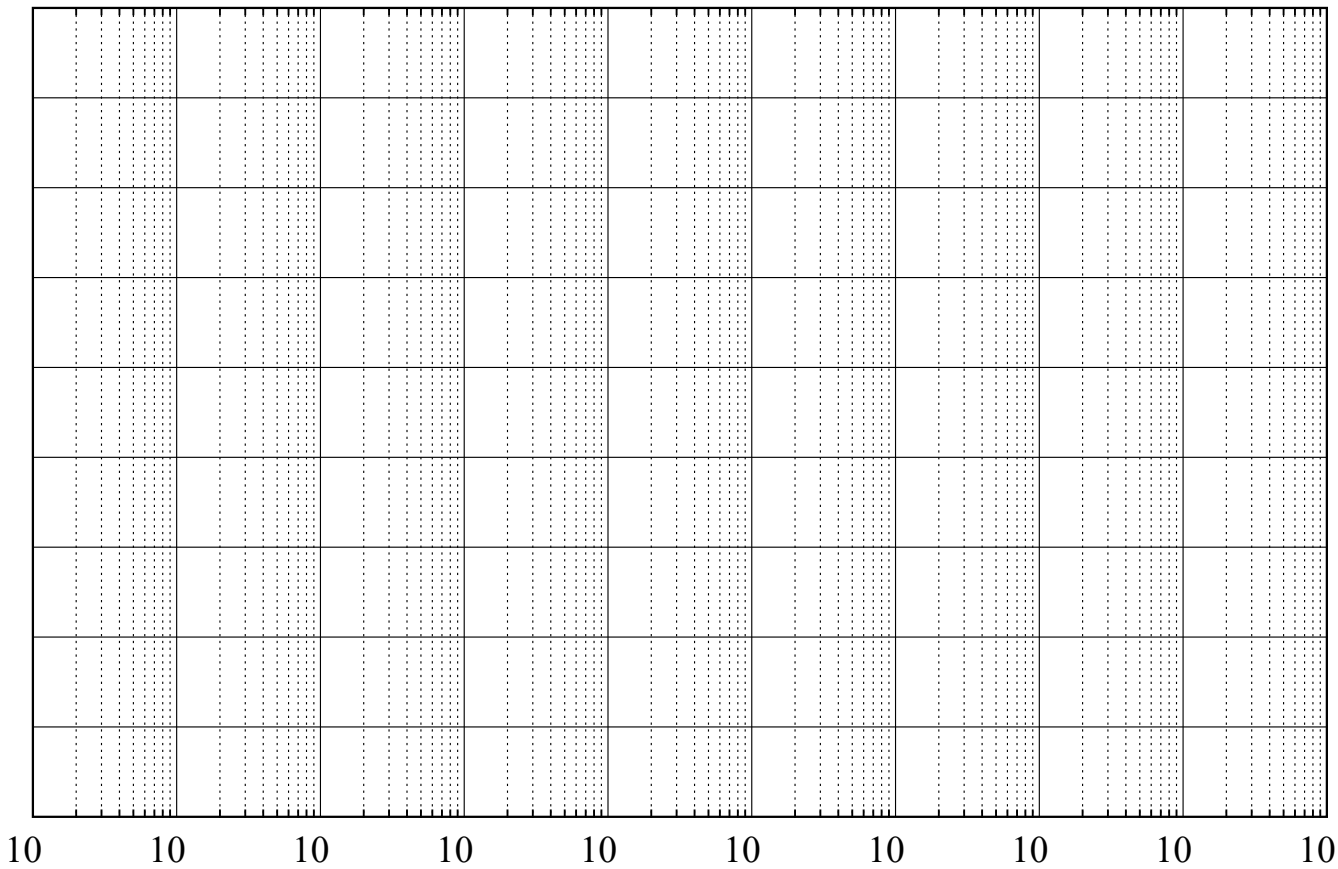

100
80
60
40
20
0
-20
-40
-60
A_0
$A_0\beta_0$
β_0
-20
-40
-60
0dB
$20\log(\beta_0) = 80$
$\rightarrow \beta = 10^{-\frac{5}{2}}$
$\beta = 0{,}00316$
$\frac{R_1}{R_1+R_2} = \beta \rightarrow R_2 = 315{,}46\ k\Omega$
10^3
10^4
10^5
10^6
10^7
10^8
10^9

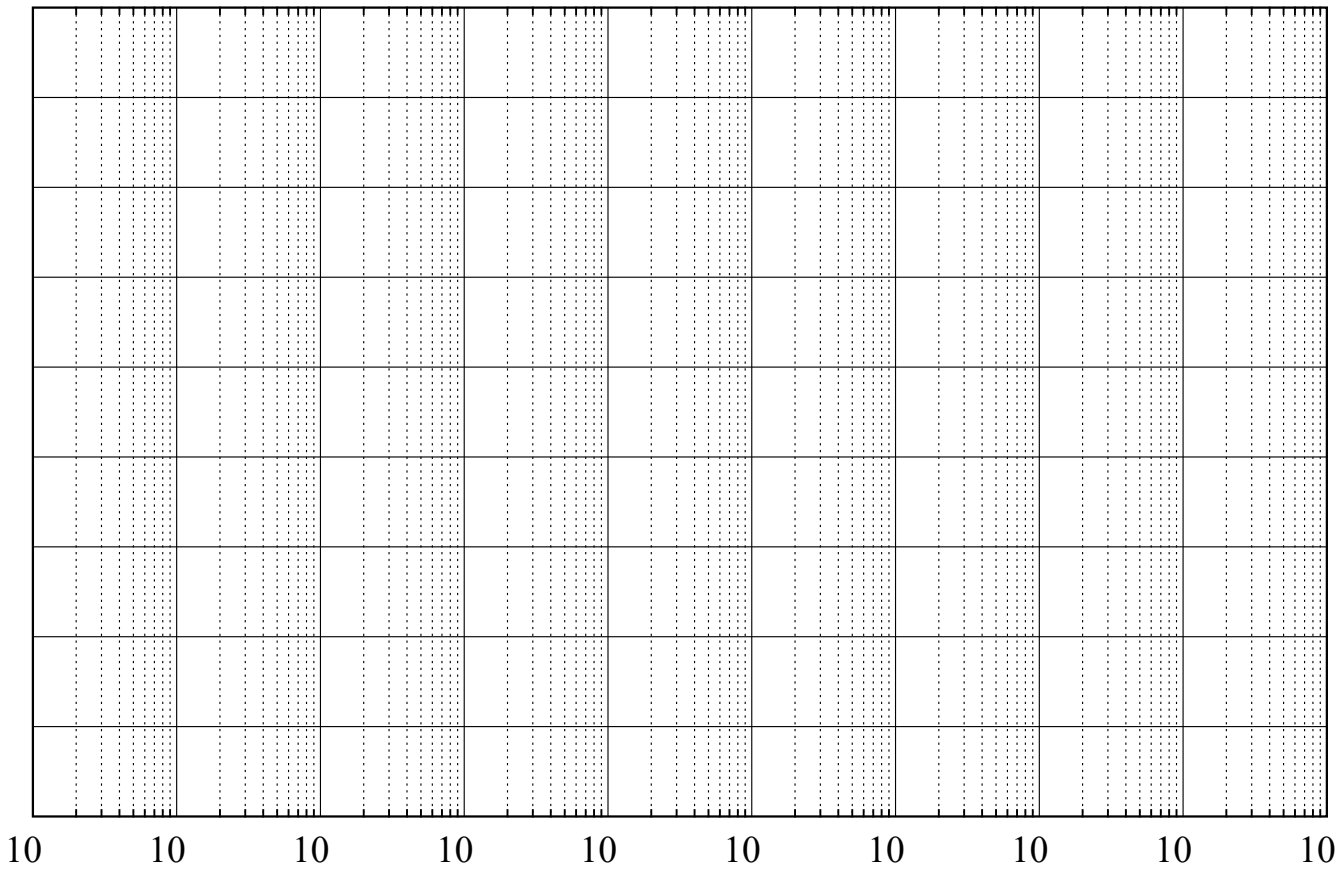

0
-45°
-90°
-180°
-270°
-360°
F.O. ≈ 68°
10
10
10
10
10
10
10
10
10
10

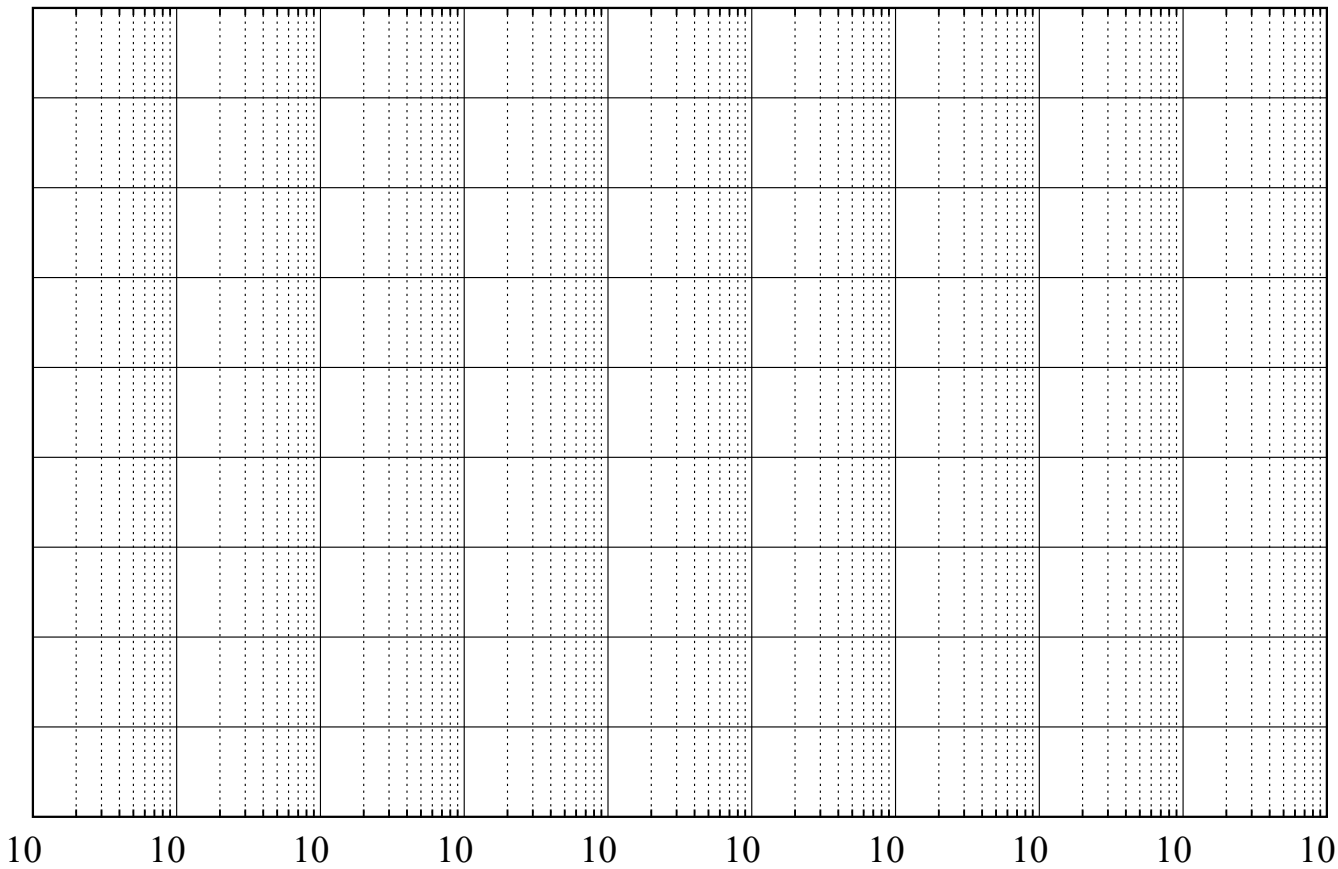

10
10
10
10
10
10
10
10
10
10


10
10
10
10
10
10
10
10
10
10


10
10
10
10
10
10
10
10
10
10


10
10
10
10
10
10
10
10
10
10


10
10
10
10
10
10
10
10
10
10


10
10
10
10
10
10
10
10
10
10